

EUNICE YIU

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Research Profile

I study how humans and AI systems build, generalize, and update models of the world, and what happens when those systems interact at scale with real people. My work combines behavioral experiments with controlled model evaluation to probe causal reasoning, interactive learning and generalization in both humans and large multimodal models. I am increasingly focused on the societal consequences of deployed AI: how interaction patterns shape learning outcomes, and how to evaluate genuine reasoning beyond surface compliance.

Employment

Sept 2025 - present **University of California, Berkeley, Berkeley, CA**
Postdoctoral Scholar, Human & Multimodal Model Evaluation affiliated with Berkeley AI Research & Department of Psychology
Active projects: causal learning and active exploration in humans and LLMs (with Google, Mila); effects of AI on human cognitive and behavioral health (with Brown ARIA)

Education

Aug 2020 – Aug 2025 **University of California, Berkeley, Berkeley, CA**
Ph.D. in Psychology
Dissertation: Relational Reasoning in Children and Machines: Insights into Causal Generalization and Innovation
Committee: Alison Gopnik, Jitendra Malik, Steve Piantadosi & Shiry Ginosar
Aug 2016 – May 2020 **Cornell University, Ithaca, NY**
B.A. in Psychology, Biological Sciences, and Economics; Cumulative GPA: 3.996/4.0

Programming Skills

Python (statistical analysis, computational modeling, PyTorch model training and evaluation, simulation environments), R, MATLAB, JavaScript / jsPsych, HTML/CSS

Grants & Fellowships

2026 **Princeton Natural & Artificial Minds Diverse Intelligences Fellowship** (\$5,000)
2025-2026 **Canadian Institute for Advanced Research Next Generation Trainee Fellowship for the Learning in Machines & Brains program** (\$10,000 CAD)
2025-2026 **7th Google - Berkeley Artificial Intelligence Research Commons Grant** (\$10,000 GCP credits)
Title: Teaching Causal Tool Use to Vision–Language Models with Human Development Data
Fine-tuning VLMs via iterative feedback loops in simulated environments and comparing their learning trajectories to human developmental patterns.
2025 **Departmental Semester Fellowship at UC Berkeley** (\$18,750)
2024-2025 **6th Google - Berkeley Artificial Intelligence Research Commons Grant** (\$41,000)
Title: Learning and Optimizing Causal Structures through Intrinsic Objectives: A Comparative Study of Human and Artificial Agents
2023-2024 **5th Meta - Berkeley Artificial Intelligence Research Commons Grant** (\$35,000)
Title: A Curriculum for Foundational AI Models Inspired by Human Cognition

Publications (* equal contribution, † undergraduate mentee)

1. **Yiu, E.**, Kosoy, E., & Gopnik, A. (2024). Transmission Versus Truth, Imitation Versus Innovation: What Children Can Do That Large Language and Language-and-Vision Models Cannot (Yet). *Perspectives on Psychological Science*, 17456916231201401.
2. **Yiu, E.**, Qraitem, M., Wong, C.†, Majhi, A. N.†, Bai, Y., Ginosar, S., ... & Saenko, K. (2025). KiVA: Kid-inspired visual analogies for testing large multimodal models. In *International Conference on Learning Representations*.
3. **Yiu, E.**, Allen, K.R., Ginosar, S., & Gopnik, A. (*in press*). Empowerment Gain and Causal Model Construction: Children and adults are sensitive to controllability and variability in their causal generalization and interventions. *Philosophical Transactions of the Royal Society A*. (Special Issue: World models, A(G)I, and the Hard problem(s) of life–mind continuity).
4. Dahmani, A.*, **Yiu, E.***, & Gopnik, A. (2025). Children Spontaneously Design Curricula to Tackle Challenging Tasks. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).
5. **Yiu, E.***, Sandbrink, K. J.*, & Gopnik, A. (2024). To observe or to bet? Investigating purely exploratory and purely exploitative actions in children, adults, and computational models. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 46).

Under Review:

1. Samiei, M.*, **Yiu, E.***, Chen, G. C., Lin, D., Richards, B., ... & Precup, D. Human adults and LLMs as Scientists: Who Benefits from Active Exploration?
2. Cheng, S., Wu, A., **Yiu, E.**, & Artzi, Y. REMAP: Evaluating Geometric Dual Representations in Multispatial Reasoning.

Invited Talks & Interviews

May 2026	AI Capabilities and Alignment Consensus Project Podcast
Jan 2026	Keynote at the Foundations of Agentic Systems Theory Workshop at AAIL, Singapore
March 2025	The Nature of Intelligence Workshop, Santa Fe Institute
June 2024	AI, Psychology & Neuroscience Summer Cluster, Simons Institute for the Theory of Computing
May 2024	Brain Science and Large Language Models Symposium, Leopoldina and Max Planck Institute for Brain Research, Germany (media coverage: <i>Frankfurter Allgemeine Zeitung</i>)
Invited Lab Talks	Stanford (Judith Fan, Michael Frank), Harvard (Tomer Ullman), Brown (Daphna Buchsbaum), UC San Diego (Caren Walker)

Professional Service

Reviewer

Cognitive Science	Journal of Experimental Psychology, Topics in Cognitive Science, Cognitive Science Society (CogSci), Society for Philosophy & Psychology (SPP), Budapest CEU Conference on Cognitive Development (BCCCD)
Artificial Intelligence	Association for Computational Linguistics (ACL), Conference on Language Modeling (COLM), Conference on Computer Vision and Pattern Recognition (CVPR)

Co-Organized Workshops and Challenges

2026	Humans of Generative AI, Conference Workshop at Computer Vision Pattern Recognition
2026	AI, Psychology and Neuroscience Summer Cluster at Simons Institute for the Theory of Computing
2026	A Unified Account of Motivation in Development, Workshop at Cognitive Development Society
2025	Kid-inspired Visual Analogies (KiVA) Challenge (multi-month competition) at Google DeepMind's Third Perception Test Workshop, International Conference on Computer Vision
2024	AI & Cognitive Development, Preconference Workshop at Cognitive Development Society